

CHEM 408 - Introduction to Green Chemistry - Fall 2025

Course Syllabus

Asynchronous Online Lecture

Instructor – **Dr. Sarah Prescott** (she/her)

Communication and Support

Best way to reach me – Send me a message through the Canvas inbox. This keeps everything in one place so I can respond quickly and not lose track.

Response Time - I typically reply the same day during weekdays (M–F, 9am–5pm ET) or by the next business day if you message me in the evening or on a weekend.

Email address - sarah.prescott@unh.edu

Office Hours - Wednesdays noon-1pm via Teams and other times by appointment

My office is Room 567 on campus, and I also hold office hours via Teams. Message me to arrange another time if you need to meet and are not able to make the office hour time.

Course Materials

We will use the annotation tool Perusall in this course, which has our course text embedded within it and is also where you will complete your problem-solving assignments. More details are provided in the course orientation.

Why "Green Goggles?" (aka About this Course)

In this course, you'll explore chemistry through a fresh set of lenses—the "green goggles" of sustainability. We'll dive into the science behind how we can design products and processes that are better for people, communities, and the planet. You'll see how chemistry connects to renewable energy, clean water, safer materials, climate solutions, and more.

You'll find plenty here to challenge your thinking, build your skills, and connect classroom concepts to real-world problems and solutions. Along the way, you'll work in interactive online spaces, share ideas with your classmates, and learn how to see the world through the same "green goggles" that scientists, engineers, and innovators use every day. Read on for more details!

Important!

Chemistry is everywhere — in the air we breathe, the water we drink, and the products we use every day. In this course, we'll use green chemistry to look at that everyday world in new ways. My hope is that you leave this class feeling accomplished, challenged, and excited about learning more. However, your safety and well-being are more important than this class. If you are facing any medical, family, housing, or food security challenges know that you can reach out and have support available (see links later in the syllabus). If you are comfortable, I hope that you will reach out to me as well and I will help you with anything I can.

My Promises

- I'm on your team - I want you to be successful and am willing to help you figure out what you need and I care about your goals.
- I will give you the tools and space to learn. I work very hard to figure out and understand the ways people learn best, embed those ways and structures in all my courses, and give you time to learn. I don't ask you to do things that do not generate a better understanding and connection to your learning. It also means that I challenge you to LEARN and not just perform tasks for a grade.
- I will actively listen and encourage you to give me feedback and communicate openly. I make changes to my courses every time I teach them in response to my own reflection along with student feedback and ideas.

We'll be learning together this semester. I'll bring the structure, content, and support — you bring your curiosity, effort, and questions.

Course Catalog Description

CHEM 408 (M1) - Green Goggles - Introduction to Green Chemistry

Term: Fall 2025 - Full Term (08/25/2025 - 12/08/2025)

Credits: 4.0

CRN: 13557

Grade Mode: Letter Grading

Description: In this course, we investigate the principles and practice of Green Chemistry as they connect to real-world examples. Green Chemistry is the field of science that uses a principle-based approach to design chemical reactions and processes to make them more sustainable. In exploring green chemistry, many of the fundamental concepts of a general chemistry course are investigated. Some topics include use of renewable feedstocks, atom economy, catalysis, waste prevention, and design for degradation.

Equivalent(s): CHEM 444G

Discovery Course

The faculty aspire for students to contextualize their major fields of study within the broader contexts of the world's challenges and opportunities. The Discovery Program, UNH's undergraduate core curriculum, supports that outcome by ensuring that as part of their bachelor's degrees, all students understand and explore a breadth of disciplinary content areas. Please note that this course meets the Physical Science Discovery Category requirement.

Student Learning Outcomes

Upon successful completion of this course, students will have the opportunity to: (*italicized items also meet outcomes for the Physical Science Discovery Category*).

- *Discuss and analyze the everyday chemistry occurring around them*
 - *Demonstrate understanding of fundamental chemical principles using the principles and practice of Green Chemistry*
 - Understand the integration of the principles and practice of Green Chemistry in a variety of chemical applications
 - Disseminate discovery of the world through “green goggles” to the larger community
 - *Continue to become more scientifically literate and use computational thinking to understand chemistry concepts and how they apply to a real-world context.*
 - Analyze scientific research and report on specific findings relevant to a larger non-scientific community, using digital and information literacy skills.
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Grading

Scale

Final grades in this course will be based on the following scale: 94-100=A; 90-93.9=A-; 87-89.9=B+; 83-86.9=B; 80-82.9=B-; 76-79=C+; 73-75=C; 70-72.9=C-; 66-69.9=D+; 63-65.9=D; 60-62.9=D-; Below 59.9=F.

Distribution

You can access your grades at any time via Canvas/Perusall. Below is a percentage distribution of all components toward your final course grade. An outline of each component is included in this syllabus. See course modules for more detailed information on each component. Please check your grades regularly and reach out to me with any questions. The distribution of main course components is highlighted below. See the course site for further details.

Perusall Assignments	30%
Discussion Boards	30%
Quizzes	30%
Final Assessment	10%

Course Structure/Format

Asynchronous Online Format

This course is fully online and asynchronous, which means we won't meet at a set time each week — instead, you'll move through interactive modules at your own pace within each week's deadlines. You'll read, watch, problem-solve, and connect with your classmates in online spaces like Perusall and our discussion boards. This format gives you flexibility, but it also means you'll need to plan ahead, stay organized, and check in regularly so you don't fall behind. I've designed the course to

guide you step-by-step, and I'm here to answer questions, clarify concepts, and help you stay on track.

To get the most out of this course, you'll need **reliable access to a desktop or laptop computer** and a **steady internet connection**. While a phone or tablet is great for quick checks, it won't be enough for completing assignments — some activities require tools or formatting that work best on a full computer. You'll also want a **scientific calculator** that can handle exponents and scientific notation. We'll use Canvas to keep everything organized, Perusall for reading and problem-solving, and occasionally Teams for meetings or questions. If you're new to any of these tools, don't worry — you'll find step-by-step guides in the orientation module, and I'm here to help you get comfortable with them early in the semester.

Staying connected each week — both with the content and with me — will make your experience more engaging and help you get the most from the course.

Course Assignments/Assessments

General Notes

- You may print out assignments and course materials for your use.
- An outline of assignments is below, but more details on all course assignments are contained on the course site.
- All assignments must be solely your own work. The use of AI-assisted writing (such as ChatGPT) is strictly prohibited on any assignment in this course unless otherwise instructed.
- Late assignments do not receive credit unless you have made arrangements with me ahead of time. That said -- please reach out if you are falling behind -- I understand that life happens and will do my best to work with you.

Perusall - Assigned Reading and Problem posing and solving - 30%

- You will use Perusall to complete assigned text reading and problem-solving activities in weekly assignments
- Enhances your learning experience, and allows you to engage and learn with others in a supportive learning environment
- Overall average score used for course grade - see Perusall site throughout the course for your individual assignment and running average scores
 - You must pass this course component overall with a 60% or higher score to pass the course.

Discussion Boards -Community Engagement and Learning

Application/Reflection - 30%

- Assignments will allow you to apply what you are learning in the course and share your discoveries with your peers
- Weekly entries will also serve to help you reflect upon your learning in the course
- Used to evaluate formative understanding of concepts in the course

- Evaluated for grading periodically
 - Late submissions will reduce the grade.

Quizzes – 30%

- Evaluate your understanding of key concepts presented in the course and give feedback on your learning.
- Two attempts on the first quiz only (highest score recorded), all others a single attempt.
- Lowest quiz score will automatically be dropped from computing overall course grade.
- Must be completed in one sitting.

Final Assessment-10%

- A final assessment with student choice to give the opportunity to research and synthesize your learning.
 - Designed to evaluate your summary understanding of fundamental course concepts.
 - Multiple options to choose from to demonstrate your synthesis of course content and ability to apply the content
 - All options require accurate content, proper formatting, clear communication, fair use, and connection to green and general chemistry knowledge
 - Additional details will be provided, and it will be open for submission following mid semester. You are encouraged to submit this portion of the exam as early as you like.
 - You must pass this course component overall with a 60% or higher score to pass the course.
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Course Policies

Class Cancellation

As this is an asynchronous course, if UNH Manchester cancels classes, it does not affect this course. Due dates will remain in effect regardless of the face-to-face courses' status.

Academic Conduct

You are encouraged to share with classmates and others the discoveries you are making and collaborative learning is built into the format of this course. However, when you submit answers to any assignments, make sure all answers are in your own words and express your own thoughts. The use of AI-assisted writing (such as ChatGPT) is strictly prohibited on any assignment in this course unless otherwise instructed.

Submitting another's work as your own is plagiarism and may result in course failure and possible further administrative action. Please ask me if you have any questions.

Courtesy and respect towards your classmates and the instructor will always be required. All online discussions must be done in a courteous and respectful manner.

Course Technology and Requirements

You'll need:

- A desktop/laptop computer with a reliable internet connection
- A scientific calculator that can perform calculations using exponents and scientific notation.
- Ability to use Google Docs/Sheets, various file formats (.doc, .docx, .pdf), web navigation, and screenshots

We will use:

- **Canvas** for course announcements, Learning Plans, Assignments/Assessments, and formal course communications (grades, accommodations, absences, etc.)
- **Perusall** for all required text readings and problem-solving assignments (orientation provided in Week 1)
- **Teams** as an optional way to connect with me for questions or office hours

Tech Tip: If you wait until just before the deadline to submit work, you take on the risk of technical issues. Always save backups — I recommend Google Drive or other auto-save cloud tools. Drafting in Google Docs before pasting into Canvas or Perusall is a good habit.

Fall 2025 Schedule

The schedule below is subject to change at instructor's discretion.

Week	Lecture Topic	Week	Lecture Topic
1	Intro to Green Chemistry	9	Bio- and Geo-spheres
2	Atoms and Elements	10	Organic and Bio-Chemistry
3	Compounds Part 1	11	Energy
4	Compounds Part 2	12	GC in your life!

5	Reactions Part 1	13	Industrial Ecology
6	Reactions Part 2	14	Thanksgiving
7	Water, solvents, solutions	15	Toxicology
8	Air and other gases		

Additional Policies, Resources, and Information

Extended Absences for Non-Academic Reasons

If you are dealing with an unexpected, extenuating circumstance that will keep you out of class or affect your performance for more than a day or two, reach out to Lisa Enright, Assistant Dean of Student Success, at lisa.enright@unh.edu to request a letter be sent to all your faculty.

If you are required to miss significant class time, you will be provided with temporary academic support so that you can continue to make satisfactory progress in this course. Please get in touch with me to discuss the specific types of support that will be implemented during your absence.

Student Wellness Support- Mental Health Services

In partnership with The Mental Health Center of Greater Manchester, UNH Manchester offers consultation visits on campus and virtually during the academic year with a mental health counselor Monday-Thursday. Students can make counseling appointments directly through an online scheduling system located on the [UNH Manchester Wellness page](#).

Services include:

- Free confidential screening & consultation with a licensed mental health therapist.
- Referrals to mental health or substance misuse treatment.
- And assistance in understanding how to afford additional treatment (with or without insurance!) or find free services.

Basic Needs Resources:

UNH has many resources that can help students who are experiencing food insecurity, housing or other financial concerns. All enrolled students are eligible for Basic Needs resources and services. A complete listing of university and local basic needs resources can be found on the [UNH Manchester Basic Needs webpage](#). The following UNH resources students are referred to most frequently are listed below:

- **The Student Emergency Financial Assistance Fund (SEFA)** provides UNH undergraduate and graduate students unable to meet essential expenses due to a temporary or unexpected hardship with short-term financial assistance once per academic year. *Please note that funding is currently limited due to high demand and UNH had to **pause funding for graduate students at this time**.* We encourage graduate students to schedule a time to meet with the Basic Needs team to discuss their options and connect them to additional resources at basic.needs@unh.edu.

UNH received a grant from Granite Edvance to support our Student Emergency Financial Assistance program specifically for undergraduate students.

To request SEFA students must complete the [Student Emergency Financial Assistance Request Form](#)

- **UNH Manchester Wildcat Cupboard**, located on the second floor, Room 290, is available to all UNH Manchester undergraduate and graduate students, staff or faculty who need access to free, nourishing food. There are an assortment of items including pasta, canned food, cereal, milk, eggs, fresh fruits and vegetables, which are purchased from the NH Food Bank through a grant or donated by members of the community. The Cupboard is open Tuesdays and Thursdays from 12:30-3:30 pm and Wednesdays from 3:00-6:00 pm. Walk ins are welcomed and no appointments are needed.
- **The UNH Manchester Wildcat Wardrobe** provides students with access to new and gently used professional attire for interviews, career fairs, networking events, and the workplace. Students can visit the Wildcat Wardrobe, located on the second floor in Room 288 (next to the Wildcat Cupboard) Monday-Friday from 9-5 pm. Items are completely free.

Confidentiality and Mandatory Reporting

The University of New Hampshire and its faculty are committed to assuring a safe and productive educational environment for all students and for the university as a whole. To this end, the university requires faculty members to report to the university's Title IX Coordinator (Bo Zaryckyj, Bo.Zaryckyj@unh.edu, 603-862-2930/1527 TTY) any incidents of sexual violence and harassment shared by students. All employees are subject to this mandatory reporting. If faculty or staff wish report an incident while on campus they can speak with Manchester's Title IX Deputy Intake Coordinator and Assistant Dean Lisa Enright 603-641-4336. Lisa's office is located on the second floor in the Library and Engagement Center, Room 236.

If a student wishes to speak to a confidential support service provider who does not have this reporting responsibility because their discussions with clients are subject to legal privilege, here is a list of confidential service providers. Including a statement of confidentiality and mandatory reporting along with links of supports for students is important for helping our community stay in compliance with Title IX, as well as living up to our commitment of building a safe community for everyone:

- [The SHARPP Center for Interpersonal Violence Awareness, Prevention, and Advocacy](#) at (603) 862-7233/TTY (800) 735-2964. SHARPP's Expanded Services Coordinator, Caroline Young, is available to meet virtually year-round Mondays-Fridays 9am-4pm and is on Manchester's campus Mondays during the Fall and Spring semesters (in Room 471). If you'd like to book a confidential appointment with her [schedule an appointment here](#). SHARPP Advocates are also available 24/7 via our Crisis Line: 603-862-SAFE (7233).
- [REACH Crisis Services](#): 24/7 Support Line: 603-668-2299. SMS/Text (M-F 8:30 AM-5:00 PM): 603-836-9620.

- For more information about what happens when a student reports, how the university treats your information once a report is made to the Title IX Coordinator, your rights and reporting options at UNH (including anonymous reporting options) [please visit student reporting options](#).
- The [uSafeUS](#) app is also available for students to keep reporting options and resources easily accessible on their phones.

Help us improve our campus and community climate. If you have observed or experienced an incident of bias, discrimination or harassment, please report the incident by contacting the Civil Rights & Equity Office at UNH.civilrights@unh.edu or TEL # (603) 862-2930 voice/ (603) 862-1527 TTY /

NameCoach

We have a new tool in Canvas, NameCoach, that will help us all to pronounce each other's names properly. Please join me in making our learning environment as inclusive as possible by recording your name (instructions [here](#)) and taking advantage of the opportunity to respectfully learn how to pronounce each other's names.

Needs of Students of all Faiths

In the event that you need accommodation for a religious or cultural holiday/observance, please let me know as early as possible.

Early Alert Progress Report

The University is invested in your academic success, and myWildcatSuccess is a tool to help you, your faculty, and your advisors communicate. If I am concerned about your academic behavior or performance, I may submit an academic alert -- particularly around Week 5. Academic alerts **are not** punitive. The goal is to provide you with support and resources to support your success. If you receive an academic alert, you and your advisor will receive an email. Alternately, find your Progress Reports under the Reports tab in myWildcatSuccess.

If you receive an alert, please meet with me or your advisor about the concern. The purpose of these alerts is to encourage you to visit the campus resources you need early on, so that you can get back on track for a successful semester.

Netiquette Guidelines

Netiquette is the socially and professionally acceptable way to communicate on the Internet. We are all expected to abide by these guidelines of "netiquette" when using online communication tools with your classmates and instructor. Guidelines can be found through the [Teaching and Learning Resource Hub](#) and at this [Academic Technology resource](#). More general guidance can be found [hereLinks to an external site.](#)

Academic Honesty

The Biology Program at UNH Manchester will strictly adhere to the University policy on academic honesty, as published in the UNH Student Rights, Rules, and Responsibilities Handbook (<http://unh.edu/vpsas/handbook/academic-honesty>). By turning in any piece of work in this course, you declare that you have read and understood the policy and that you did not engage in any form of academic dishonesty as defined in the Handbook.

Plagiarism can take many forms, such as: submitting someone else's work - in whole or in part - as your own; collaborating on answers for individual assignments or allowing your own work to be used

by another student; copying information from a website or other text without proper documentation; buying a pre-written paper or lab report.

Cheating is mainly concerned with copying on exams or in the lab, bringing crib notes into an exam, or referring to notes or the textbook or any other source such as a programmable calculator, tablet, or cell phone during an exam.

Any instances of cheating or plagiarism will result in consequences that can range from a failing grade on the assignment for all students involved to dismissal from the University, as defined in the UNH Student Rights, Rules, and Responsibilities Handbook.

Any form of academic misconduct on any graded work in this course (including assignments, quizzes, or exams) will result in an automatic zero on the assignment. Additional consequences may result at the discretion of the instructor.

Credit Hour Compliance

This syllabus reflects the federal definition of a credit hour, which entails a minimum 3 hours of engaged time per week per credit over a 15-week semester. Examples of engaged time include class time, assignments, examinations, laboratories, participation in course-related experiences (attending a talk or performance, speakers and events, fieldwork, etc.), conferences, and office hours. Student work reflects intended learning outcomes and is verified through evidence of student achievement. For more information, please see: [NECHE policy on Credits and Degrees.Links to an external site.](#)

Class Recording

Class lectures and activities will/may be recorded in order to make this course more accessible, and recordings may be made available to students presently enrolled in the class to assist in reviewing material and preparing for assessments. Recordings are to be used for educational use/purposes only. Students may not record class activities (including audio/photographs/screenshots of instructors and students) without express permission from their instructor and students whose image is captured. If permission is granted, student-initiated recordings may only be used for educational purposes in this class and only during the period in which the student is enrolled in the class; authorized student-initiated recordings may not be posted or shared in any fashion outside of the class.

Accessibility Services

According to the Americans with Disabilities Act (as amended, 2008), each student with a disability has the right to request services from UNH to accommodate his/her/their disability. If you are a student with a documented disability or believe you may have a disability that requires accommodations, please contact Student Accessibility Services (SAS) located on the Manchester campus in room 417 or sas.office@unh.edu.

Accommodation letters are created by SAS with the student. Please follow-up with your instructor as soon as possible to ensure timely implementation of the identified accommodations in the letter. Faculty have an obligation to respond once they receive official notice of accommodations from SAS but are under no obligation to provide retroactive accommodations.

For more information refer to www.unh.edu/sas or contact SAS at 603.862.2607, 711 (Relay NH) or sas.office@unh.edu.

Library

The UNH Manchester librarians are available to assist you with the research process. Visit our website at <https://cps.unh.edu/libraryLinks to an external site.> to explore library services and search for

reliable academic sources. You can also contact the library at 603-641-4173 or unhm.library@unh.edu.

Helpful links for students:

- Make a Research Appointment: <https://libraryguides.unh.edu/remotearchive/researchhelpLinks to an external site.>
- Library Search Box: https://libraryguides.unh.edu/librarysearchbox_unhmanchesterLinks to an external site.
- Reserve a Study Room: <https://cps.unh.edu/library/support-servicesLinks to an external site.>
- Citing Sources: <https://libraryguides.unh.edu/unhmcitingsourcesLinks to an external site.>
- Evaluating Sources: <https://libraryguides.unh.edu/ENGL401UNHManchester/evaluatingresourcesLinks to an external site.>

UNH provides a variety of resources to support students' academic success, health and well-being, and engagement with the campus community.

A list of student resources can be found on [the Provost's website](#). These resources are not required for this course but may be helpful to you during the semester.

Knack Peer Tutoring

Knack is a peer-to-peer tutoring platform that is available to all enrolled students for all undergraduate courses in Durham and Manchester at no cost to students. Students looking for additional assistance outside of the classroom are advised to consider working with a peer tutor through Knack. UNH has partnered with Knack to provide students with access to verified tutors who have successfully completed your course. To view available tutors, visit unh.joinknack.comLinks to an external site. and sign in with your student account.

Questions about Knack Tutoring can be sent to Stephanie Kirylych, Director of Advising, at stephanie.kirylych@unh.edu.

Connors Writing Center and Online Writing Lab

Connors Writing Center (CWC)

The [Connors Writing CenterLinks to an external site.](#) offers one-on-one writing conferences to current UNH students (undergraduate and graduate). We work with writers from all disciplines on many different kinds of writing. Our conferences are conducted by trained writing assistants who are UNH undergraduate and graduate students.

The CWC offers both in-person and virtual appointments, available to any current UNH student. The online sessions are synchronous (real-time) appointments designed to help students have one-on-one conversations about their writing at any stage of the process, mirroring what we do during in-person appointments. [Students can schedule an appointment Links to an external site.](#) any time during the semester.

Online Writing Lab (OWL)

The [Online Writing Lab \(OWL\) Links to an external site.](#) assists students via asynchronous remote collaboration with qualified writing assistants. The OWL is not an editing or manuscript preparation service. Feedback may include reference to mechanics on the submitted paper, but it will be targeted to bring attention to patterns and issues in order to help the writer self-correct the paper and carry forward lessons to improve future writing.

Students in all fields of study can submit up to 20 double-spaced pages of any form of writing, including course papers, resumes, creative essays, or personal statements. Within 3 business days, writing assistants provide feedback that can be printed or viewed on screen. The OWL is accessible through any computer that has an Internet connection and Microsoft Word. To submit a paper, students can sign in at [unh.mywconline.comLinks to an external site.](#) and select the "Online Writing Lab" schedule.

Should you have any questions about the policies or information in this syllabus, please contact me to discuss them. By continuing in this course, you agree to the policies outlined here. This syllabus is subject to change at my discretion, and any updates will be communicated in Canvas.